YEAR 8 MATHS · CHAPTER 1 · MIND MAP

Computation with integers

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1 Adding and subtracting

Line up the place values, then carry or borrow. Rounding first can make it easier in your head.

$347 + 586 = 933 \quad 502 - 178 = 324$

In your head: $198 + 57 = 200 + 57 - 2 = 255$

Estimate first: $347 + 586 \approx 350 + 590 = 940$

2 Multiplying and dividing

Split the number into parts you can multiply, then add the parts.

$34 \times 26 = 34 \times 20 + 34 \times 6 = 680 + 204 = 884$

$756 \div 7 = 108$

3 Number properties

Order doesn't matter for + and ×: $a + b = b + a$. Brackets can be expanded: $a(b + c) = ab + ac$.

$7 \times 23 = 7 \times 20 + 7 \times 3 = 161$

12 and 18: HCF is 6, LCM is 36

4 Divisibility and prime factorisation

Divisible by 2: even. By 3: digits add to a multiple of 3. By 5: ends in 0 or 5. By 9: digits add to a multiple of 9.

Keep dividing by primes: $60 = 2^2 \times 3 \times 5$ **and** $84 = 2^2 \times 3 \times 7$

1 is not a prime number. 2 is the only even prime.

5 Negative integers

Numbers get smaller as you move left on the number line.

$-5 < -2 \quad -1 > -8$

A temperature of −3°C is colder than 2°C

In order: $-8, -3, 0, 2$

A rise of 5° from $-3°$C gives $2°$C

6 Adding and subtracting negatives

Adding a negative means subtract. Subtracting a negative means add.

$-3 + 7 = 4 \quad 5 + (-8) = -3$

$-2 - (-6) = -2 + 6 = 4$

$-4 - 3 = -7$

7 Multiplying and dividing negatives

Same signs give a positive. Different signs give a negative.

$-4 \times 6 = -24 \quad -36 \div (-9) = 4$

$-2 \times (-3) \times (-1) = -6$: an odd number of negatives gives a negative

Watch out: $(-3)^2 = 9$ but $-3^2 = -9$

8 Order of operations and substitution

Brackets, then powers, then × and ÷ from left to right, then + and − from left to right.

$2 + 3 \times (-4) = 2 - 12 = -10$

If $a = -2$ **and** $b = 3$: $a^2 - b = 4 - 3 = 1$

Watch out: put negative numbers in brackets when you substitute.

YEAR 8 MATHS · CHAPTER 2 · MIND MAP

Angles and geometrical figures

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1 Reviewing angles

Complementary angles add to 90°. Supplementary angles add to 180°. Vertically opposite angles are equal.

35° has a complement of **55°** and a supplement of **145°**

Angles on a straight line add to 180°

Angles around a point add to 360°

Acute: under 90°. Right: 90°. Obtuse: 90° to 180°. Reflex: over 180°.

2 Transversal and parallel lines

With parallel lines: corresponding angles are equal (F), alternate angles are equal (Z), co-interior angles add to 180° (C).

If one angle is 70°: corresponding **70°**, alternate **70°**, co-interior **110°**

Watch out: these rules only work when the lines are parallel.

3 Triangles

Angles add to 180°. An exterior angle equals the two opposite interior angles added together.

Angles 40° and 75°: the third is **65°**, the exterior angle is **115°**

Equilateral: all angles 60°

Isosceles: two equal sides, two equal base angles

4 Quadrilaterals

Angles add to 360°.

Angles 80°, 95°, 110°: the fourth is **75°**

Parallelogram: opposite angles equal

Rhombus: all sides equal

Kite: one pair of opposite angles equal

5 Polygons

Angle sum of an n-sided polygon: $S = (n-2) \times 180°$.

Pentagon: $3 \times 180° = 540°$, each angle of a regular one is $108°$

Hexagon: $4 \times 180° = 720°$, each angle of a regular one is $120°$

6 3D solids and Euler's formula

For solids with flat faces: $F + V = E + 2$ (faces, vertices, edges).

Cube: $6 + 8 = 12 + 2$ ✓

Triangular prism: $5 + 6 = 9 + 2$ ✓

Prisms have two matching ends. Pyramids come to a point.

7 3D coordinates

A point is (x, y, z): along, across, then up. The origin is $(0, 0, 0)$.

$(2, 3, 4)$ is 2 along the x-axis, 3 along the y-axis and 4 up the z-axis

A 5 by 3 by 2 box with a corner at the origin has its opposite corner at $(5, 3, 2)$

YEAR 8 MATHS · CHAPTER 3 · MIND MAP

Fractions, decimals and percentages

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1 Equivalent fractions

Multiply or divide the top and bottom by the same number.

$\frac{2}{3} = \frac{8}{12}$ $\frac{18}{24} = \frac{3}{4}$

2 Operations with fractions

Add or subtract: common denominator first. Divide: flip the second fraction and multiply.

$\frac{2}{3} + \frac{1}{4} = \frac{8}{12} + \frac{3}{12} = \frac{11}{12}$

$1\frac{1}{2} \times \frac{2}{3} = \frac{3}{2} \times \frac{2}{3} = 1$ $\frac{3}{4} \div \frac{1}{2} = \frac{3}{2}$

3 Negative fractions

The same sign rules as integers.

$-\frac{1}{2} + \frac{1}{3} = -\frac{3}{6} + \frac{2}{6} = -\frac{1}{6}$

$-\frac{2}{3} \times \left(-\frac{3}{4}\right) = \frac{1}{2}$

4 Understanding decimals

Compare place by place, starting from the left.

0.305 is 3 tenths and 5 thousandths

0.5 is bigger than 0.45: 5 tenths beats 4 tenths

Watch out: more digits doesn't mean bigger.

5 Operations with decimals

Line up the decimal points to add or subtract. To multiply, count the decimal places. To divide, make the divisor whole.

$2.75 + 1.4 = 4.15$

$0.3 \times 0.2 = 0.06$ (two decimal places)

$4.8 \div 0.6 = 48 \div 6 = 8$

6 Recurring decimals and rounding

Divide the top by the bottom. It either stops (terminating) or repeats (recurring).

$\frac{3}{8} = 0.375$ $\frac{1}{3} = 0.\dot{3}$ $\frac{2}{11} = 0.\dot{1}\dot{8}$

2.467 to 1 decimal place is **2.5**

7 Converting

Fraction to decimal: divide. Decimal to percentage: multiply by 100.

$\frac{3}{5} = 0.6 = 60\%$

$\frac{1}{8} = 0.125 = 12.5\%$

$7\% = 0.07 = \frac{7}{100}$

8 Finding percentages

Percentage of an amount: multiply. Express as a percentage: divide and multiply by 100.

25% of 80 is **20**

18 out of 40: $\frac{18}{40} \times 100\% = 45\%$

9 Increasing and decreasing

Use a multiplier: 100% plus the increase, or 100% minus the decrease.

Up 15%: $60 \times 1.15 = 69$

Down 20%: $45 \times 0.8 = 36$

10 Percentage change, profit and loss

$$\%\ \text{change} = \frac{\text{change}}{\text{original}} \times 100\%$$

Bought for \$50, sold for \$65: profit \$15, $\frac{15}{50} = 30\%$

11 The unitary method

Find what 1% is worth, then multiply to get what you need.

15% of an amount is \$60

1% is $60 \div 15 = \$4$, so 100% is $\$400$

Check: 15% of \$400 is \$60 ✓

YEAR 8 MATHS · CHAPTER 4 · MIND MAP

Measurement and Pythagoras

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1 Length and perimeter

Perimeter is the distance around the outside.

1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm

Rectangle 8 by 3: $P = 2(8 + 3) = 22$

2 Circumference

$C = \pi d = 2\pi r$

$d = 10$: $C = 10\pi \approx 31.4$

$r = 3$: $C = 6\pi \approx 18.8$

3 Area

Rectangle lw. Triangle $\frac{1}{2}bh$.

$6 \times 4 = 24$ $\quad \frac{1}{2} \times 8 \times 5 = 20$

1 m² = 10 000 cm²

4 Special quadrilaterals

Parallelogram bh. Rhombus or kite $\frac{1}{2}xy$. Trapezium $\frac{1}{2}(a + b)h$.

Trapezium 4, 10, height 5: 35

Rhombus, diagonals 6 and 8: 24

5 Area of circles

$A = \pi r^2$. Use the radius, half the diameter.

$r = 5$: $A = 25\pi \approx 78.5$

$d = 8$: $r = 4$, so $A = 16\pi \approx 50.3$

Watch out: πr^2 is area, $2\pi r$ is circumference.

6 Sectors and composite shapes

Sector: $\frac{\theta}{360} \times \pi r^2$. Composite: split into shapes you know, then add or take away.

Quarter circle $r = 4$: $\frac{1}{4} \times 16\pi = 4\pi \approx 12.6$

8 by 4 rectangle with a semicircle on the 4 side: $32 + \frac{1}{2}\pi(2^2) \approx 38.3$

7 Surface area of prisms

Add the areas of every face. Drawing the net helps.

Box 5 by 3 by 2: $2(15 + 10 + 6) = 62$

8 Volume and capacity

Box: $V = lwh$.

$5 \times 3 \times 2 = 30$ cm³

1 cm³ = 1 mL, 1000 cm³ = 1 L, 1 m³ = 1000 L

9 Volume of prisms and cylinders

$V = Ah$. Cylinder: $V = \pi r^2 h$.

Triangular prism: $\frac{1}{2} \times 3 \times 4 = 6$, length 10: $V = 60$

Cylinder $r = 2,\ h = 5$: $V = 20\pi \approx 62.8$

10 Time and time zones

Going west, clocks are earlier. Going east, clocks are later.

Melbourne is UTC+10, Perth is UTC+8

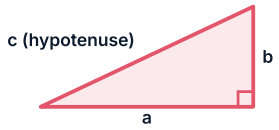
3 pm in Melbourne is **1 pm** in Perth (standard time)

Watch out: daylight saving changes the gap in summer.

11 Pythagoras' theorem

Right-angled triangles: $c^2 = a^2 + b^2$, c is the hypotenuse.

Sides 5 and 12: $c^2 = 25 + 144 = 169$, so $c = 13$



12 Finding a shorter side

Subtract: $a^2 = c^2 - b^2$.

Hypotenuse 10, side 6: $a^2 = 100 - 36 = 64$, so $a = 8$

Watch out: a shorter side can never be longer than the hypotenuse.

YEAR 8 MATHS · CHAPTER 5 · MIND MAP

Algebra and index laws

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1 The language of algebra

In $3x + 5$: x is a pronumeral, 3 is the coefficient, 5 is the constant.

"4 less than twice n" is $2n - 4$

2 Substitution and equivalence

Replace each letter with its value. Equivalent expressions are equal for every value.

$x = 3,\ y = -2$: $2x + y = 4 \quad xy = -6$

$2(x + 1)$ is equivalent to $2x + 2$

3 Adding and subtracting terms

Only like terms (same letters, same powers) can be added or subtracted.

$5a + 2b - a = 4a + 2b$

$3x^2 + x - x^2 = 2x^2 + x$

4 Multiplying and dividing terms

Multiply the numbers, then the letters. Divide like a fraction and cancel.

$3a \times 4b = 12ab \quad 2x \times 5x = 10x^2$

$12ab \div 3a = 4b$

Watch out: $x + x = 2x$ but $x \times x = x^2$

5 Algebraic fractions: + and −

Use a common denominator, then combine the numerators.

$\frac{x}{2} + \frac{x}{3} = \frac{3x}{6} + \frac{2x}{6} = \frac{5x}{6}$

$\frac{3a}{4} - \frac{a}{2} = \frac{3a}{4} - \frac{2a}{4} = \frac{a}{4}$

6 Algebraic fractions: × and ÷

Cancel common factors, then multiply. To divide, flip the second fraction.

$\frac{2a}{5} \times \frac{10}{a} = 4$

$\frac{a}{4} \div \frac{a}{8} = \frac{a}{4} \times \frac{8}{a} = 2$

7 Expanding brackets

Multiply every term inside by the term outside.

$3(x + 4) = 3x + 12$

$-2(x - 5) = -2x + 10$

$x(x + 3) = x^2 + 3x$

8 Factorising

The opposite of expanding: take out the highest common factor.

$6x + 9 = 3(2x + 3)$

$x^2 + 5x = x(x + 5)$

$12ab - 8a = 4a(3b - 2)$

9 Applying algebra

Use a letter for the unknown and write a rule.

Phone plan: \$20 plus \$0.50 a text: $C = 20 + 0.5t$. **40 texts:** \$40

Rectangle, width x, length $x + 3$: perimeter $4x + 6$

10 Index laws: × and ÷

Same base: multiply, add the indices. Divide, subtract them.

$x^3 \times x^4 = x^7 \quad a^6 \div a^2 = a^4$

$2^3 \times 2^2 = 2^5 = 32$

Watch out: $2^3 \times 2^2$ is not 4^5.

11 Zero index and power of a power

Anything to the power 0 is 1. Power of a power: multiply the indices.

$5^0 = 1 \quad 3x^0 = 3 \quad (x^2)^3 = x^6$

YEAR 8 MATHS · CHAPTER 6 · MIND MAP

Ratios and rates

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1 Simplifying ratios

Divide both parts by the HCF. Use the same units first.

$12 : 18 = 2 : 3$

50 cm : 2 m $= 50 : 200 = 1 : 4$

$0.5 : 2 = 1 : 4$

Three parts: $6 : 9 : 15 = 2 : 3 : 5$

Watch out: convert to the same units before you simplify.

2 Solving ratio problems

Add the parts, find one part, then multiply.

$60 in the ratio 2 : 3: 5 parts, 1 part = $12

So **$24** and **$36**

In the ratio 3 : 5 with the first part 12: the second is 20

$90 in the ratio 1 : 2 : 3: 6 parts, 1 part = $15, so **$15**, **$30** and **$45**

3 Scale drawings

A scale of 1 : 100 means 1 cm on the drawing is 100 cm in real life.

1 : 100, 4.5 cm on the plan: $4.5 \times 100 = 450$ cm, or 4.5 m

1 : 50 000, 3 cm on the map: 150 000 cm, or 1.5 km

Finding the scale: a 6 m wall drawn 3 cm long is 3 cm : 600 cm = **1 : 200**

Watch out: use the same units on both sides of the scale.

4 Introducing rates

A rate compares two different units. Read "per" as "for each".

150 km in 3 hours: **50 km/h**

$12 for 4 kg: **$3/kg**

Best buy: 2 L for $5 is $2.50/L; 3 L for $6.60 is $2.20/L, so the **3 L** is better value

5 Solving rate problems

Multiply the rate by the time, or divide to find the time.

8 L/min for 30 min: $8 \times 30 = 240$ L

200 L at 8 L/min: $200 \div 8 = 25$ min

Typing 45 words a minute for 20 minutes: $45 \times 20 = 900$ words

Check the units match before you multiply or divide.

6 Speed

$S = \frac{D}{T}, D = S \times T, T = \frac{D}{S}$

240 km at 80 km/h: $T = 3$ h

90 km in 1.5 h: $S = 60$ km/h

45 minutes at 80 km/h: $D = 80 \times 0.75 = 60$ km

72 km/h in m/s: $72 \div 3.6 = 20$ m/s

Watch out: 45 minutes is 0.75 hours, not 0.45.

7 The unitary method

Find the value of one, then multiply to get what you need.

5 pens cost $8: 1 pen is $1.60, so 12 pens are $\$19.20$

In a ratio, 3 parts = 45: 1 part is 15, so 7 parts are 105

A 250 g pack costs $4.50: 100 g costs $4.50 \div 2.5 = \$1.80$

Finding the value of one works for any amount.

YEAR 8 MATHS · CHAPTER 7 · MIND MAP

Equations and inequalities

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1 Reviewing equations

An equation has an equals sign. The solution is the value that makes it true.

$x + 7 = 12 \Rightarrow x = 5$ $3x = 21 \Rightarrow x = 7$

Check by substituting: $5 + 7 = 12$ ✓

2 Equivalent equations

Do the same to both sides. Undo in reverse order.

$2x + 3 = 11 \Rightarrow 2x = 8 \Rightarrow x = 4$

3 Equations with fractions

Clear the fraction by multiplying both sides.

$\frac{x}{3} = 5 \Rightarrow x = 15$

$\frac{x+1}{2} = 4 \Rightarrow x + 1 = 8 \Rightarrow x = 7$

$\frac{x}{4} + 1 = 3 \Rightarrow \frac{x}{4} = 2 \Rightarrow x = 8$

4 Pronumeral on both sides

Subtract the smaller term so the pronumeral is on one side.

$5x + 2 = 3x + 10 \Rightarrow 2x = 8 \Rightarrow x = 4$

5 Equations with brackets

Expand the brackets first.

$3(x - 2) = 12 \Rightarrow 3x - 6 = 12 \Rightarrow x = 6$

$2(x + 1) = x + 6 \Rightarrow 2x + 2 = x + 6 \Rightarrow x = 4$

6 Simple quadratic equations

Get x^2 by itself, then take the square root. Remember both answers.

$x^2 = 25 \Rightarrow x = \pm 5$

$x^2 + 3 = 19 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$

$x^2 = -4$ has no solution: a square can't be negative.

7 Formulas and relationships

Substitute what you know, then solve for what's left.

$P = 2l + 2w$ with $l = 5,\ w = 3$: $P = 16$

$A = lw$ with $A = 24,\ w = 4$: $l = 6$

8 Applications of equations

Let a letter be the unknown, write an equation, solve it, answer in words.

Double a number and add 5 to get 17: $2x + 5 = 17$, so the number is 6

Rectangle, width w, length $2w$, perimeter 30: $6w = 30$, so 5 by 10

9 Inequalities

$<$ less than, $\leq$ less than or equal, $>$ greater than, $\geq$ greater than or equal.

$x > 2$: open circle at 2, arrow right

$x \leq 4$: closed dot at 4, arrow left

10 Solving inequalities

Solve like an equation, but flip the sign when you multiply or divide by a negative.

$2x + 1 < 9 \Rightarrow 2x < 8 \Rightarrow x < 4$

$-3x \leq 12 \Rightarrow x \geq -4$

Watch out: flip the sign when dividing by a negative.

YEAR 8 MATHS · CHAPTER 8 · MIND MAP

Probability and statistics

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1 Interpreting graphs and tables

Read the title, the axes, the scale and the key first.

Column graphs for categories, line graphs for change over time, pie charts for parts of a whole

2 Frequency tables

Tally each result, then count. Frequency is how many times it happened.

5 out of 20 results: relative frequency $\frac{5}{20} = 25\%$

3 Graphs of frequency tables

Histograms show numerical data: the columns touch.

Also: dot plots, and stem-and-leaf plots (key 2 | 4 = 24)

4 Measures of centre

Mean: add and divide. Median: middle value in order. Mode: most common.

1, 3, 3, 6, 7: mean $\frac{20}{5} = 4$, median 3, mode 3

2, 4, 6, 9: median $\frac{4+6}{2} = 5$

5 Measures of spread

Range = max − min. IQR $= Q_3 - Q_1$, the spread of the middle half.

2, 4, 5, 7, 8, 10: range 8

$Q_1 = 4,\ Q_3 = 8$, so IQR $= 4$

Split the data in half at the median, then find the median of each half.

6 Surveying and sampling

A sample should represent the population. Random samples avoid bias.

Census: everyone. Sample: some

Asking only your friends isn't a random sample

7 Probability

$P = \frac{\text{favourable}}{\text{total}}$, from 0 (impossible) to 1 (certain).

One die: $P(5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$

$P(\text{not } 6) = 1 - \frac{1}{6} = \frac{5}{6}$

8 Two-step experiments

List every outcome in a table.

Two coins: HH, HT, TH, TT.

$P(\text{one head}) = \frac{2}{4} = \frac{1}{2}$

Two dice: $P(\text{sum of } 12) = \frac{1}{36}$

9 Tree diagrams

Multiply along the branches, then add the results you want.

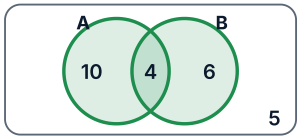
3 red and 1 blue, with replacement:

$P(\text{red, red}) = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$

10 Venn diagrams and two-way tables

Fill in the overlap first.

25 people: $P(\text{both}) = \frac{4}{25}$



11 Experimental probability

Relative frequency $= \frac{\text{times it happened}}{\text{number of trials}}$

18 heads in 50 tosses: $\frac{18}{50} = 0.36$. The expected probability is 0.5.

More trials usually get closer to the expected probability.

YEAR 8 MATHS · CHAPTER 9 · MIND MAP

Linear relationships

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1 The Cartesian plane

(x, y): across first, then up or down. The origin is $(0, 0)$.

$(-2, 3)$ is 2 left and 3 up, in the second quadrant

2 Rules, tables and graphs

Use the rule to fill a table, plot the points and join them.

$y = 2x + 1$: $x = 0, 1, 2$ gives $y = 1, 3, 5$

3 Finding the rule

The step in y is the number in front of x. The value at $x = 0$ is the constant.

x: 0, 1, 2, 3 and y: 4, 7, 10, 13

Up 3 each time, starts at 4: $y = 3x + 4$

4 Solving equations with graphs

Read across from the value you want to the line, then down to the x-axis.

Solve $2x + 1 = 7$ on $y = 2x + 1$: at $y = 7$, $x = 3$

5 Solving inequalities with graphs

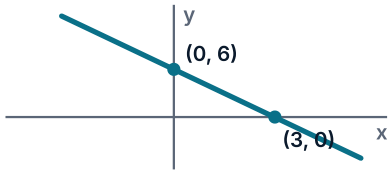
Find the part of the line above or below the value you want.

$2x + 1 < 7$: the line is below $y = 7$ when $x < 3$

6 The x- and y-intercepts

x-intercept: put $y = 0$. y-intercept: put $x = 0$.

$2x + y = 6$: x-intercept $(3, 0)$, y-intercept $(0, 6)$



7 Gradient

Gradient $= \frac{\text{rise}}{\text{run}}$. Up to the right is positive, down is negative.

$(1, 2)$ to $(3, 6)$: $\frac{4}{2} = 2$

Horizontal lines have a gradient of 0

8 Gradient–intercept form

$y = mx + c$: m is the gradient, c is the y-intercept.

$y = 3x - 2$: $m = 3,\ c = -2$

$y = -x + 4$: $m = -1,\ c = 4$

9 Applications of straight lines

c is the starting value, m is the rate.

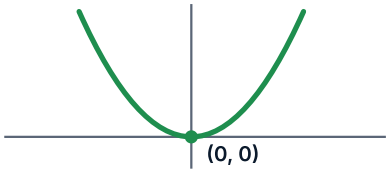
Taxi: \$3 plus \$2 per km: $C = 2d + 3$

5 km: $C = 2 \times 5 + 3 = \$13$

10 Non-linear graphs

Not every rule gives a straight line. $y = x^2$ is a parabola.

$y = x^2$: $(-2, 4), (0, 0), (2, 4)$



YEAR 8 MATHS · CHAPTER 10 · MIND MAP

Transformations and congruence

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1 Reflection

A mirror image: each point is the same distance from the mirror line.

$(2, 3)$ in the x-axis: $(2, -3)$

$(2, 3)$ in the y-axis: $(-2, 3)$

2 Translation

A slide, with no turning or flipping.

Right 3, down 2: $(1, 4) \to (4, 2)$

3 Rotation

A turn, given by a centre, an angle and a direction.

180° about the origin: $(2, 3) \to (-2, -3)$

90° clockwise about the origin: $(2, 3) \to (3, -2)$

4 Congruent figures

Exactly the same shape and size ($\cong$). Reflections, translations and rotations keep figures congruent.

Matching sides and matching angles are equal

Write the vertices in matching order

5 Congruent triangles

Tests: SSS, SAS, AAS, RHS.

SSS: three sides

SAS: two sides and the angle between them

AAS: two angles and a side

RHS: right angle, hypotenuse, side

Watch out: AAA isn't a test: same angles can be different sizes.

6 Tessellations

Shapes that cover a surface with no gaps or overlaps. The angles at each point add to 360°.

Regular hexagons: $3 \times 120^\circ = 360^\circ$ ✓

Regular pentagons don't: 108° doesn't divide evenly into 360°

7 Congruence and quadrilaterals

A diagonal can split a quadrilateral into congruent triangles, which proves its properties.

A parallelogram's diagonal makes two congruent triangles, so opposite sides are equal

A rhombus's diagonals cross at right angles

8 Similar figures

Same shape, different size. Scale factor $= \dfrac{\text{image length}}{\text{original length}}$.

3 cm becomes 12 cm: scale factor 4

The angles stay the same

9 Similar triangles

Same angles, sides in the same ratio. Use the scale factor to find missing sides.

Matching sides 3 and 12: scale factor 4, so side 5 matches 20

A 1 m stick has a 2 m shadow; a tree's shadow is 10 m: the tree is 5 m tall